Section 3

Identifying the of Bacterial Species

1. Cultural Characteristics

Cultural characteristics are great importance for the identification and classification of microbes and clarify the microbial diversity.

Cultivation experiments of microorganisms include the visible appearance to the naked eye of growth over or in different media. The aim of studying the culture characteristics of bacterial colonies is to identify the different types of bacteria.

A. <u>Colonial shape</u>

- Circular
- Filamentous
- Irregular
- Drastic
- Spindle





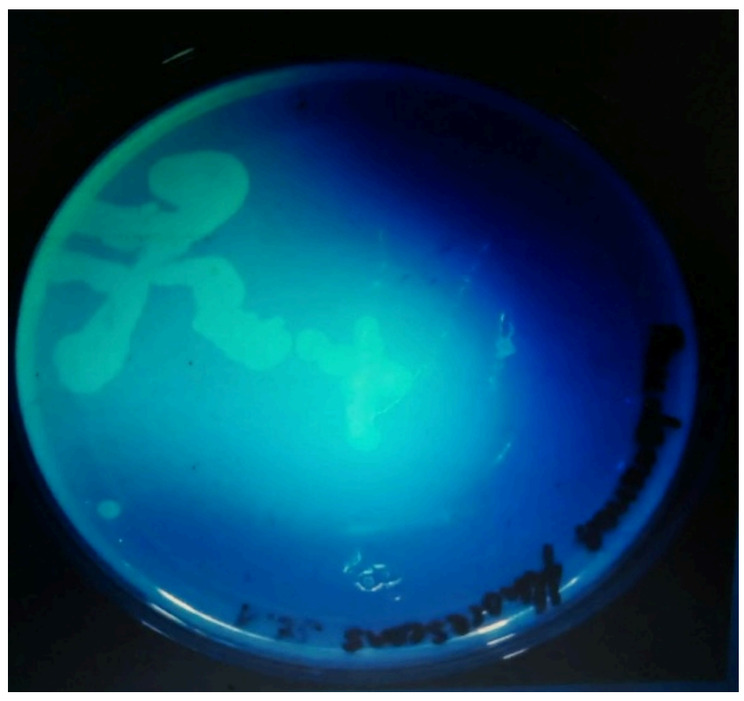
B. Colonial color

1. Some bacteria secrete dyes that are not soluble in water. The color of the colonies may be yellow, red, .etc.

Examples are *Xanthomonas* bacteria whose colonies appear in yellow.



2. Some other bacteria also secrete water-soluble dyes that color the medium and do not color the colonies, such as some species of the genus *Pseudomonas*, which secrete a water-soluble fluorescein dye that has another yellowish color.



C. Colony’s elevation

- Flat
- High
- Convex
- Many convexities
- High center



D. Colony’s edge shape

•Complete

• Corrugated

• Lobed

• Serrated

• Wrinkled

• Filamentous



E. Colony’s surface

- Smooth
- Rough

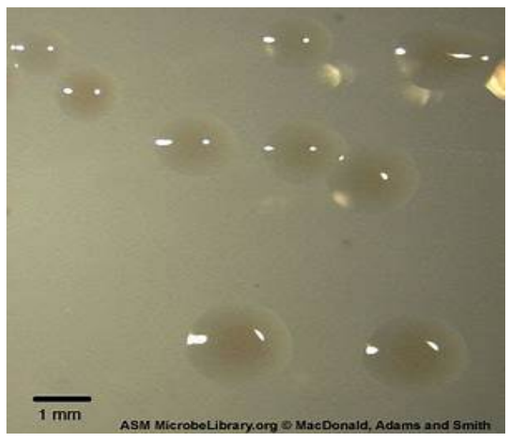
F. The optical characteristics of the colony

- **Opaque**: Do not allow light to pass through.

- **Translucent**: Allows light to pass through but does not allow objects to be fully seen behind.

G. Colony strength

- Zebedee
- Sticky
- Membranous
- Fragile



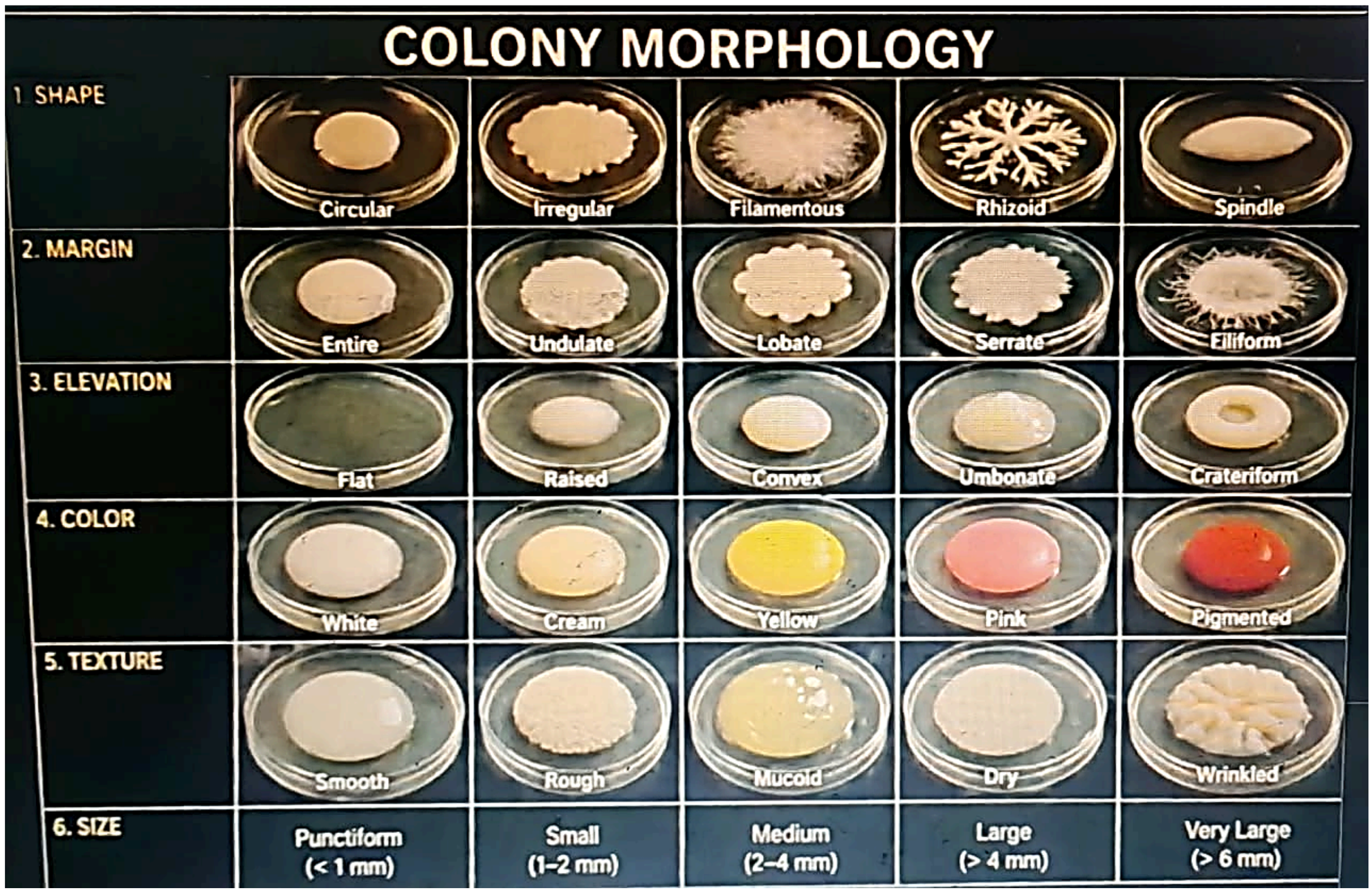


H. The smell

- It has no smell
- It has smell.

MARGIN	COLOUR	ELEVATION	TEXTURE	SHAPE
Curled	Orange	Raised	Slimy, moist	Round
Entire (smooth)	Red or pink	Umbonate	Matte, brittle	Punctiform
Filamentous	Black	Flat	Shiny, viscous	Rhizoid (root-like)
Undulate (wavy)	Brown	Convex	Dry, mucoid	Filamentous
Lobate	Opaque or white	Pulvinate (Cushion-shaped)	Translucent	Irregular
Erose (serrated)	**Milky**	**Growth into culture medium**	**Iridescent (changes colour in reflected light)**	**Spindle**

Source: MicroDok


COLONY MORPHOLOGY
1. SHAPE
Circular
Irregular
Filamentous
Rhizoid
Spindle
2. MARGIN
Entire
Undulate
Lobate
Serrate
Elliform
3. ELEVATION
Flat
Raised
Convex
Umbonate
Crateriform
4. COLOR
White
Cream
Yellow
Pink
Pigmented
5. TEXTURE
Smooth
Rough
Mucoid
Dry
Wrinkled
6. SIZE
Punctiform (< 1 mm)
Small (1–2 mm)
Medium (2–4 mm)
Large (> 4 mm)
Very Large (> 6 mm)

2. Morphological Characteristics

The morphological examination of bacterial cells can be carried out in two ways:

A. Examination of living cells without staining:

Living bacterial cells are examined without staining to observe their natural characteristics, especially their movement (motility).

B. Examination of dead cells with staining:

Bacterial cells are fixed and stained with suitable dyes to make their shape and structural details easier to observe under the microscope.

the main advantages of the dye are:

1. Increase the contrast between microorganisms and the background in which they are, allowing to distinguish between different morphological types.

2. Facilitate the study of the internal structures of bacterial cells, such as: cell walls, capsule, and spores.

3. Enable the researcher to use a higher magnification power.